

# A Discrete-Sine-Transform based Preconditioner for Adaptive Meshes

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## Abstract

We present a preconditioner for solving fractional differential equations (FDEs) on an adaptive mesh. Adaptive refinement of the problem domain results in a stiffness matrix with Toeplitz blocks along the main diagonal, while the FDE yields a dense stiffness matrix where off-diagonal blocks are stored as low-rank approximations. Our adaptive preconditioner utilizes ideas from the circulant preconditioner of Chan and Strang [SIAM Journal on Scientific Computing, 1989] and the discrete-sine-transform based preconditioner of Bini and Di Benedetto [Association for Computing Machinery, 1990]. We take advantage of the Toeplitz blocks on the diagonal and accounts for the low-rank nature of the off-diagonal blocks. We demonstrate the adaptive preconditioner's effectiveness at accelerating convergence for our systems and emphasize its efficient application. This work presents theoretical results about the spectral clustering of the preconditioned system. In order to prove these results, special consideration is taken on how the low-rank blocks perturb the eigenvalues of the Toeplitz block-diagonal system. Numerical tests are used to inspect any assumptions and validate our results.

Preconditioner, Adaptive Refinement, Finite-element Method, Toeplitz, Circulant, DST

## 1 Introduction

In this paper we consider fractional differential equations (FDEs) of the form

$$D_x^\alpha u(x) = f(x), \quad x \in (a, b), \quad (1)$$

where  $D_x^\alpha$  is the Riesz fractional derivative of order  $1 < \alpha < 2$  and  $f \in L^2([a, b])$ .

FDE applications... HOWEVER, (next paragraph)

FDEs present unique difficulties in numerical methods due to the non-locality of the operator resulting in a dense stiffness matrix. Additionally, as part of the fractional derivative the singular kernel,  $k(x, s) = |x - s|^{1-\alpha}$ , is integrated, resulting in end-point singularities which make Galerkin finite-element methods sub-optimal [9, 12]. To address this, Jin and Zhou developed a reconstruction technique that isolates the singularity to improve FEM convergence [7]. Alternatively, Wang and Du developed a Toeplitz preconditioner that reduces iteration cost to  $\mathcal{O}(n \log n)$  using the preconditioned conjugate gradient squared method [12].

The above methods require uniform meshes. On a uniform mesh the FEM discretization results in a stiffness matrix that is Toeplitz. All methods on a uniform mesh leverage this structure for efficient linear solves. However, these methods fail to address high error near singularities, and instead the size of the mesh must grow. An optimal mesh would have the  $L^\infty$  error equally distributed on each interval, but a uniform mesh does not, in general, satisfy this condition. Indeed, the singularities of FDEs force higher error in the intervals near the boundary and thus require very fine meshes to achieve uniform error below a given threshold. Alternatively, to achieve desired accuracy across the whole domain we use an adaptive mesh and adaptive finite-element method (AFEM) [10, 13]. The usual AFEM algorithm iterates through the process

$$\text{SOLVE} \rightarrow \text{ESTIMATE} \rightarrow \text{MARK} \rightarrow \text{REFINE}, \quad (2)$$

but the computational cost is dominated by the **SOLVE** step [13].

Our aim is to build a preconditioner for the stiffness matrix,  $A$ , from an AFEM discretization of Equation (1) that will reduce this work. We investigate what structure the adaptive mesh retains and how that is reflected in the structure of  $A$ . Then, using preconditioning techniques formulated for Toeplitz matrices [4, 5, 3, 6], we build a preconditioner, show the preconditioned system has eigenvalues clustered around 1, and demonstrate numerical results with superlinear convergence.

## 2 Background

### 2.1 Properties of Structured Matrices

There has long been interest in solving Toeplitz linear systems efficiently. A matrix  $A$  is called Toeplitz if  $a_{ij} = a_{i-j}$ . Arbitrary  $n \times n$  matrices have up to  $n^2$  entries and are solved directly in  $\mathcal{O}(n^3)$  FLOPS. Since a Toeplitz matrix has at most  $2n - 1$  unique entries, we expect to solve Toeplitz systems with  $\mathcal{O}(n^2)$  FLOPS, (see for example Levinson's algorithm, Trench's algorithm). Even with these improvements, however, fast Toeplitz algorithms are unstable in general and infeasible for large systems that arise in application [2]. Infinite or periodic Toeplitz matrices can be solved via a transformation to Fourier space and a

discrete convolution in TODO time. On the other hand, finite Toeplitz systems are much more difficult to solve due to the boundaries of the mesh or matrix. To solve these systems, we turn to iterative Krylov and multigrid methods [6, 9, 13]. For these methods we take advantage of Toeplitz structure by using structured preconditioners.

Toeplitz matrices commonly arise in partial differential equations discretization, FDEs, signal processing, and control theory. In application, Toeplitz matrices are often symmetric positive definite (SPD) or symmetric positive semidefinite (SPSD). Given an SPD Toeplitz system  $Lx = b$ , the idea introduced by Strang in TODO (A proposal for Topelitz Matrix Calcs 1986) is to use a circulant preconditioner,  $C$ , so that  $C^{-1}Tx = C^{-1}b$  is solved in fewer iterations. A circulant matrix is a Toeplitz matrix, that additionally has the “wrap-around” property where the last entry of each row is the first entry of the subsequent row:

$$L = \begin{pmatrix} \ell_0 & \ell_{-1} & \cdots & \ell_{-(n-1)} \\ \ell_1 & \ell_0 & \cdots & \ell_{-(n-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n-1} & \ell_{n-2} & \cdots & \ell_0 \end{pmatrix} \quad C = \begin{pmatrix} c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{pmatrix}. \quad (3)$$

Equation 1:  $L$  is a Toeplitz matrix and  $C$  is circulant.

As will be discussed in Section (2.2), the stiffness matrix resulting from AFEM applied to (1) is not Toeplitz, but does contain SPD Toeplitz blocks. Toeplitz matrices have many special properties, here we focus on their connection to functions in the Wiener class. [[4]] Given a singly infinite, symmetric Toeplitz matrix  $L$  we define the finite subsection  $L_m$  to be the  $m \times m$  principle submatrix. That is,

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \cdots \\ \ell_1 & \ell_0 & \ell_1 & \ddots \\ \ell_2 & \ell_1 & \ell_0 & \ddots \\ \cdots & \ddots & \ddots & \ddots \end{pmatrix} \quad L_m = \begin{pmatrix} \ell_0 & \ell_1 & \cdots & \ell_{m-1} \\ \ell_1 & \ell_0 & \cdots & \ell_{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{m-1} & \ell_{m-2} & \cdots & \ell_0 \end{pmatrix}.$$

Assume  $\sum_{k=-\infty}^{\infty} |\ell_k| < \infty$ , then the matrices induce functions  $\tilde{\ell}(z) = \sum_{k=-\infty}^{\infty} \ell_k z^k$  and  $\tilde{\ell}_m(z) = \sum_{k=-(m-1)}^{m-1} \ell_k z^k$  are real, positive, and in the Wiener class for  $|z| = 1$ .

Previous work on Toeplitz systems offers many options for circulant preconditioners, (see [3]). Strang and R. Chan first proposed to construct a circulant  $C$  by copying the central diagonals of  $L$  into the required wrap-around entries. Assuming for simplicity  $n = 2m$ , then the first column of  $C$  would be  $\ell_0, \dots, \ell_m, \dots, \ell_0$  [4]. Alternatively, Tony Chan constructed a circulant precon-

ditioner,  $C$ , by averaging the diagonals of  $L$ . He shows this to be optimal in the sense that  $\|C - L\|_F$  is minimal. TODO CITE

Although their construction differs, they are all diagonalized by some trigonometric transform matrix and are thus efficiently applied via the fast Fourier transform (FFT). The spectrum of the preconditioned systems are asymptotically the same [3]. In this paper the preconditioner we construct is based on the  $\tau$  matrix class, introduced in [1].

The  $\tau$  preconditioner is not itself circulant, but is diagonalized by the type-I discrete sine transform (DST) matrix. Kailath and Olshevsky used a displacement operator kernel framework to unify the circulant preconditioners,  $\tau$ , and other preconditioners diagonalized by other discrete trigonometric matrices. Together these form a class of discrete-trigonometric-based preconditioners. Though our construction could be generalized to circulant preconditioners, we use  $\tau$  because it does not require complex arithmetic needed for the FFT. Following [1], given a symmetric Toeplitz matrix  $L$  we can build a Hankel correction matrix,  $H$ , where

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \cdots & \ell_{n-2} & \ell_{n-1} \\ \ell_1 & \ell_0 & \ell_1 & \cdots & \ell_{n-3} & \ell_{n-2} \\ \ell_2 & \ell_1 & \ell_0 & \cdots & \ell_{n-4} & \ell_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \ell_{n-2} & \ell_{n-3} & \ell_{n-4} & \cdots & \ell_0 & \ell_1 \\ \ell_{n-1} & \ell_{n-2} & \ell_{n-3} & \cdots & \ell_1 & \ell_0 \end{pmatrix} \quad H = \begin{pmatrix} \ell_2 & \ell_3 & \ell_4 & \cdots & \ell_{n-1} & 0 & 0 \\ \ell_3 & \ell_4 & \ell_5 & \cdots & 0 & 0 & 0 \\ \ell_4 & \ell_5 & \ell_6 & \cdots & 0 & 0 & \ell_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \ell_5 & \ell_4 & \ell_3 \\ 0 & 0 & \ell_{n-1} & \cdots & \ell_4 & \ell_3 & \ell_2 \end{pmatrix}. \quad (4)$$

Then define  $\tau := L - H$ . Both  $L$  and  $H$  are symmetric, so they are completely represented by at most  $n$  unique values: the diagonals of  $L : \ell_0, \ell_1, \dots, \ell_{n-1}$ , and the anti-diagonals of  $H : \ell_2, \ell_3, \dots, \ell_{n-1}, 0, 0$ .

The following lemma summarizes properties of  $\tau$ . [see [1]] Let  $L$  be a Toeplitz matrix and  $H$  the corresponding Hankel correction as in Equation (4). Then  $\tau = L - H$  has the following properties:

1.  $\tau$  is diagonalized by the DST,  $S$ . We write  $\tau = S\Lambda S^{-1} = S\Lambda S$ .
2.  $\tau$  can be applied in  $\mathcal{O}(n \log n)$  FLOPS.
3. Let  $L_1$  be the first column of  $L$ . Then, the  $k$ -th eigenvalue of  $\tau$  is proportional to the  $k$ -th entry of  $SL_1$ . Specifically,

$$\lambda_k(\tau) = c_k [SL_1]_k, \text{ where } c_k := \sqrt{\frac{n+1}{2}} \frac{1}{\sin(\frac{\pi k}{n+1})}. \quad (5)$$

4. Each eigenvalue of  $\tau$  is written as the truncated function  $\tilde{\ell}_m$  evaluated somewhere on the unit circle,  $\lambda_k(\tau) = \tilde{\ell}_m(z_k)$ , where  $|z_k| = 1$ .

## 2.2 AFEM stiffness matrix

Again, let  $A$  be the stiffness matrix resulting from AFEM applied to (1). During the MARK module of AFEM (2), it is often the case that adjacent elements have

similar error estimates, and are either both marked or both not marked for refinement. As a result, after several refinements there are still subsections of the domain that have been refined to the same level. We refer to these subdomains as *locally uniform*. Since uniform meshes give rise to SPD Toeplitz systems, wherever there are adjacent elements of the same size we find a corresponding Toeplitz block on the main diagonal of our stiffness matrix,  $A$ . The size of this block depends on how many adjacent elements are the same size. In practice we have observed the greatest level of refinement is required near the boundary of the domain. In general we have larger Toeplitz blocks from locally uniform subdomains in the middle of the matrix, and smaller blocks—or indeed  $1 \times 1$  blocks—near the boundary. These  $1 \times 1$  blocks, along with their corresponding rows and columns, can be viewed as additional boundaries imposed onto the system. As mentioned earlier, it is the boundaries of a Toeplitz system that necessitate more computation while solving, so we can anticipate that wise handling of these  $1 \times 1$  block disruptions to the Toeplitz structure will be the key to making a good preconditioner. We use  $A_D$  to denote the block diagonal with Toeplitz blocks part of  $A$ .

The fractional derivative results in a dense matrix, but we observe off-diagonal decay, as elements that are physically separated in the domain have weaker interaction. The rate of decay depends on the fractional order  $\alpha$ . Blocks that are most distant from the diagonal contain less information and we assume they have small-norm. We denote these blocks as  $A_F$ . We call the blocks that are either directly below or to the right of the Toeplitz blocks “blocks adjacent to the diagonal”, together they are denoted  $A_E$ . These blocks are not in general small-norm, but due to the weaker interaction, are well-suited for low-rank approximation, (see [13] for more details). These assumptions are made precise in section 4.2 and numerically validated in section 5.1. Using this notation, we write our stiffness matrix as the sum of the Toeplitz blocks, blocks adjacent to the diagonal, and blocks further removed from the diagonal,  $A = A_D + A_E + A_F$  (see Figure (1)).

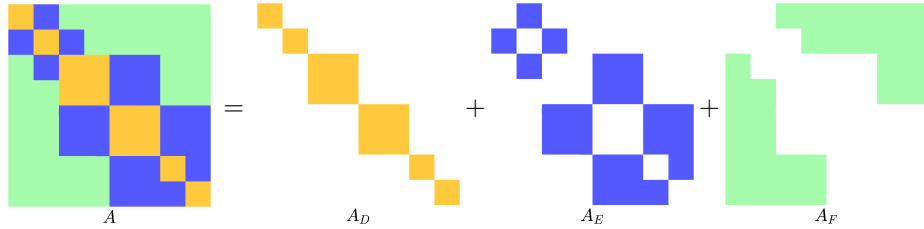


Figure 1: Illustration of the stiffness matrix splitting.  $A_D$  consists of Toeplitz blocks on the diagonal,  $A_E$  is the block to the right of and below the each Toeplitz block, and  $A_F$  is everything else.

good spot to mention Toeplitz compression RNLA solvers?

### 3 Adaptive $\tau$ tau Preconditioner

In this section we set forth the properties we require from a preconditioner, how we use  $\tau$  in building the adaptive  $\tau$  preconditioner, which we call  $\mathcal{T}$ , and how to build and apply  $\mathcal{T}$  efficiently. A good preconditioner for an iterative method must in general decrease the total number of iterations, without increasing the cost of a single iteration. We borrow Kailath's [8] criteria for preconditioners, similar criteria is well-established in the literature (for example [1]).

1. Complexity of constructing and applying  $\mathcal{T} \in \mathbb{R}^{m \times m}$  should be  $\mathcal{O}(m \log m)$ .
2. A linear system with  $\mathcal{T}$  should be solved in  $\mathcal{O}(m \log m)$  operations.
3. The spectrum of  $\mathcal{T}^{-1}A$  should be clustered around 1.

How tightly the eigenvalues cluster around 1 determines the speed of convergence. We use the infinite matrix framework established in Section (2.1). In Section (4.1), we summarize results established previously about the spectral clustering of  $\tau$  applied to a single Toeplitz block,  $L$ . This result is then used as a lemma in proving the spectral clustering for our stiffness matrix from the adaptive mesh.

To build  $\mathcal{T}$ , assume the stiffness matrix has  $k$  Toeplitz blocks  $L_1, L_2, \dots, L_k$  of respective sizes  $m_1, m_2, \dots, m_k$ . Assume additionally these are ordered as they appear along the main diagonal. Each block has a corresponding singly infinite matrix with a corresponding generating function,  $\ell_1(z), \ell_2(z), \dots, \ell_k(z)$ . Assume that for all  $1 \leq i \leq k$ ,  $\ell_i(z) \sum_{j=-\infty}^{\infty} |t_j| < \infty$  and  $\ell_i(z) > 0$  for  $z$  on the unit circle.

To explicitly construct  $\mathcal{T}$  for the adaptive system, we calculate the Hankel correction  $H_i$  for every Toeplitz block,  $L_i$  and define  $\tau_i = L_i - H_i$ . Using the matrices  $\tau_1, \tau_2, \dots, \tau_k$  we assemble  $\mathcal{T}$  as the block diagonal matrix with  $\tau_i$  as the  $i$ -th block.

$$\mathcal{T} = \begin{pmatrix} \tau_1 & & & \\ & \tau_2 & & \\ & & \ddots & \\ & & & \tau_k \end{pmatrix}.$$

This explicit construction is not the most efficient, instead we apply  $\mathcal{T}^{-1}$  implicitly in  $\mathcal{O}\left(\sum_{j=1}^k m_j \log(m_j)\right) \leq \mathcal{O}(km \log(m))$ , where  $m := \max_j m_j$ . Let  $S_{m_i}$  denote the type-I DST matrix of size  $m_i$ . Using Equation (5) to calculate the eigenvalues of  $\tau_i$ , we write  $\tau_i = S_{m_i} \Lambda_i S_{m_i}$ .

$$\mathcal{T}^{-1} = \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_k} \end{pmatrix} \begin{pmatrix} \Lambda_1^{-1} & & & \\ & \Lambda_2^{-1} & & \\ & & \ddots & \\ & & & \Lambda_m^{-1} \end{pmatrix} \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_k} \end{pmatrix}. \quad (6)$$

The DST blocks in the outside matrices can be applied via the fast sine transform (FST) to do matrix multiplication in  $\mathcal{O}(m \log m)$  FLOPS. The eigenvalue matrix is diagonal and inverted in  $m$  operations, and can be multiplied against a Toeplitz matrix embedded in a  $2m \times 2m$  circulant matrix in  $\mathcal{O}(m \log m)$  FLOPS. Moreover, both the FST and block arithmetic are highly parallelizable for more efficient implementation. This establishes Criteria (1) and (2) for  $\mathcal{T}$ .

### 3.1 Handling of $1 \times 1$ blocks

## 4 Theoretical Results

In this section we prove that the spectral clustering in criteria (3) holds for  $\mathcal{T}$ . Proofs for clustering of discrete-trigonometric-transform based preconditioned problems in this area tend to follow a similar structure which we also follow: we show the preconditioned matrix can be written as the identity matrix plus a small-norm matrix and a low-rank matrix,

$$\mathcal{T}^{-1}A = I + A_{SN} + A_{LR}.$$

The small-norm term has spectrum clustered around 0, forcing the clustering of the entire system around 1 with the number of outliers bounded by the rank of the low-rank term.

We present our version of the spectral clustering proof for the case of  $\tau$  acting on a single Toeplitz block. We then expand this result to prove the clustering of the block matrix applied to our the Toeplitz blocks and corresponding off diagonal blocks contained in the  $A_{22}$  block after permuting. Finally, we show how our handling of the  $1 \times 1$  blocks preserves the spectral clustering.

Let  $T \in \mathbb{R}^{n \times n}$  be an invertible matrix. Let  $A \in \mathbb{R}^{n \times n}$ , then

$$\lambda_k(T^{-1}A) \leq \frac{\lambda_k(A)}{\lambda_{\min}(T)}.$$

Using the min-max theorem,

$$\begin{aligned} \lambda_k(T^{-1}A) &= \min_{\dim V=k} \max_{x \in V} \left( \frac{(Ax, x)}{(Tx, x)} \right) \\ &\leq \min_{\dim V=k} \left[ \max_{x \in V} \left( \frac{(Ax, x)}{(x, x)} \right) \max_{x \in V} \left( \frac{(x, x)}{(Tx, x)} \right) \right] \\ &\leq \left[ \min_{\dim V=k} \max_{x \in V} \left( \frac{(Ax, x)}{(x, x)} \right) \right] \max_{x \in \mathbb{R}^n} \left( \frac{(x, x)}{(Tx, x)} \right) \\ &= \lambda_k(A) \max_{x \in \mathbb{R}^n} \left( \frac{(x, x)}{(Tx, x)} \right) \leq \lambda_k(A) \frac{1}{\lambda_{\min}(T)}. \end{aligned}$$

## 4.1 Spectral Clustering of a Toeplitz Block

As in other spectral clustering proofs in this field, we aim to show that  $\mathcal{T}^{-1}A$  has the form  $I + A_{SN} + A_{LR}$ , where  $I$  is the identity,  $A_{SN}$  has small  $\ell^2$ -norm, and  $A_{LR}$  has low rank. There are many proofs of this type in the literature, see for example TODO. The following lemma closely follows these proofs, our contribution is the special handling of off-diagonal blocks and the  $1 \times 1$  blocks that disrupt our structure. [Eigenvalue Clustering on a Toeplitz Block, [1, 8]] Let  $L$  be a singly infinite, symmetric, Toeplitz matrix with diagonals  $\ell_0, \ell_1, \dots$ , and  $L_m$  is its  $m \times m$  truncation with diagonals  $\ell_0, \ell_1, \dots, \ell_{m-1}$ . Assume the generating function  $\tilde{\ell}(z)$  is in the Wiener class and positive on the unit circle. Let  $\tau$  be the corresponding preconditioner, then for  $m$  large enough the spectrum of  $\tau^{-1}L$  is clustered around 1. Since  $\tilde{\ell}(z) > 0$  for  $|z| = 1$ , compactness of the unit circle implies that for some  $\varepsilon > 0$ ,  $\tilde{\ell}(z) > 2\varepsilon$  for all  $|z| = 1$ . By the Wiener class assumption we choose  $N$  such that  $\sum_{j=N}^{\infty} \ell_j z^j \leq \sum_{j=N}^{\infty} |\ell_j| < \varepsilon$ . So for all  $m \geq N$ ,

$$2\varepsilon < \ell(z) = \tilde{\ell}_m(z) + \sum_{j=m}^{\infty} \ell_j z^j < \tilde{\ell}_m(z) + \varepsilon.$$

Thus  $\tilde{\ell}_m(z) > \varepsilon$  on the unit circle.

Recall that by construction  $\tau = L - H$ , where  $H$  is the Hankel matrix with anti-diagonals  $\ell_2, \ell_3, \dots, \ell_{m-1}, 0, 0$ . So  $L = \tau + H$  and  $\tau^{-1}L = I + \tau^{-1}H$ . Therefore it suffices to show that the eigenvalues of  $\tau^{-1}H$  are clustered around 0.

We split  $H$  into a low-rank matrix  $H_{LR}$  that contains the anti-diagonals  $\ell_0, \dots, \ell_N$  and a small-norm matrix  $H_{SN}$  such that  $H = H_{LR} + H_{SN}$ . Let  $s := \text{rank}(H_{LR}) \ll m$ . The small-norm descriptor is justified since  $H_{SN} = H - H_{LR}$  is a Hermitian  $m \times m$  matrix with at most two copies of  $\ell_N, \dots, \ell_{m-1}$  in each row/column. Thus  $\|H_{SN}\|_2 = \sqrt{\|H_{SN}\|_1 \|H_{SN}\|_{\infty}} = \|H_{SN}\|_1 < 2\varepsilon$ . Hence by Weyl's Inequality at least  $m - s$  of the eigenvalues of  $H_{SN}$  are clustered within  $2\varepsilon$  of zero.

Now by Lemma (4),

$$\lambda_k(\tau^{-1}H) \leq \lambda_k(H) \frac{1}{\lambda_{\min}(\tau)} = \lambda_k(H) \frac{1}{\tilde{\ell}_m(z_{\min})} \leq \lambda_k(H) \frac{1}{\varepsilon}. \quad (7)$$

Thus, the clustering of the spectrum of  $H$  around 0 implies the same for  $\tau^{-1}H$ . Equation (7) shows that the bigger  $\lambda_{\min}(\tau)$ , the stronger the bound. We can get this eigenvalue exactly using Lemma (2.1), and equation (5). It can also be bounded in terms independent of  $\tau$ , using the generating function  $\tilde{\ell}(z)$ , as above.

## 4.2 Spectral Clustering of the Full System

Let  $A$  be the stiffness matrix as described in Section (2.2) and  $\mathcal{T}$  as described in Section (3). Then the spectrum of  $\mathcal{T}^{-1}A$  is clustered around 1.

Recall that the stiffness matrix is dense with Toeplitz blocks on the diagonal (denoted  $A_D$ ), low-rank blocks adjacent to the diagonal (denoted  $A_E$ , and small-norm blocks elsewhere (denoted  $A_F$ ), so that  $A = A_D + A_E + A_F$ . Let  $H$  be the block diagonal Hankel Correction matrix, observing then that  $A_D = \mathcal{T} + H$  we see that  $A = \mathcal{T} + H + A_E + A_F$ . Our approach is to split  $H + A_E + A_F$  into a small-norm term  $A_{SN}$  and a low-rank term  $A_{LR}$ . Thus

$$\mathcal{T}^{-1}A = \mathcal{T}^{-1}(\mathcal{T} + A_{SN} + A_{LR}) = I + \mathcal{T}^{-1}A_{SN} + \mathcal{T}^{-1}A_{LR}. \quad (8)$$

Hence, it suffices to show that the spectrum of  $\mathcal{T}^{-1}A_{SN}$  is clustered around 0 and that  $\mathcal{T}^{-1}A_{LR}$  contributes a bounded number of eigenvalue outliers.

Recall that blocks in  $A_E$  are well-approximated as a low-rank matrix. To that end, we begin by examining a single block in  $A_E$ , call it  $E$  (see Figure (1) for reference). Let  $E$  be a  $m_E \times m_E$  matrix with approximation rank  $r_E$ . Then we split  $E$  via its singular value decomposition (SVD):  $E = \left( \sum_{i=1}^{r_E} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top \right) + \left( \sum_{i=r_E+1}^{m_E} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top \right)$ . Here  $\mathbf{u}$  and  $\mathbf{v}$  are length  $m_E$  vectors, with a slight abuse of notation, we embed these in length  $m$  vectors to express the block matrix  $A_E$  as the SVD of its blocks. Let  $W$  represent the set of blocks in  $A_E$ . Define  $r := \max_{E \in W} r_E$  and  $M := \max_{E \in W} m_E$ . We write  $A_E$  as the sum of the SVD of its blocks, and then regroup terms into a low-rank sum and a small-norm sum.

$$\begin{aligned} A_E &= \sum_{E \in W} \left[ \left( \sum_{i=1}^r \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top \right) + \left( \sum_{i=r+1}^{m_E} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top \right) \right] \\ &= \left( \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \right) + \left( \sum_{i=r+1}^M \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \right). \end{aligned}$$

As before, we have  $\varepsilon > 0$  and choose  $N$  much smaller than  $m$  so that for every Toeplitz block the truncated function for the block  $a_{m_i}(z) > \varepsilon$  on the unit circle and  $\sum_{i=N}^{\infty} |a_i| < \varepsilon$ . We additionally split  $H$  by separating into  $H_{LR}$  the anti-diagonals with coefficients  $a_0, \dots, a_N$  and into  $H_{SN}$  the anti-diagonals comprising of  $a_{N+1}, \dots, a_{m_i}$  for each block. So we have two splittings,

$$A_E = \left( \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \right) + \left( \sum_{i=r+1}^M \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \right), \quad H = H_{SN} + H_{LR}.$$

The first term in each sum can be thought of as the low-rank equivalent from before and similarly the second term is the small-norm summand. Finally, we make the splitting  $A = \mathcal{T} + A_{SN} + A_{LR}$ , where

$$A_{SN} = H_{SN} + \sum_{i=r+1}^M \sigma_i \mathbf{u}_i \mathbf{v}_i^\top + A_F, \quad A_{LR} = H_{LR} + \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top.$$

Now  $s := \text{rank}(\mathcal{T}^{-1}A_{LR}) \leq \text{rank}(A_{LR}) \leq 2(N-1) + r$  bounds the number of outlier eigenvalues. It remains to show  $\|\mathcal{T}^{-1}A_{SN}\|_2 \leq \varepsilon$ . Define  $\tilde{A}_E =$

$\sum_{i=r_{A_E}+1}^M \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ , so that  $A_{SN} = H_{SN} + \tilde{A}_E + A_F$ , then

$$\|\mathcal{T}^{-1}A_{SN}\|_2 \leq \|\mathcal{T}^{-1}H_{SN}\|_2 + \|\mathcal{T}^{-1}\tilde{A}_E\|_2 + \|\mathcal{T}^{-1}A_F\|_2.$$

Since the block multiplication  $\mathcal{T}^{-1}H_{SN}$  is each  $\tau$  acting on its corresponding Hankel correction, it is bounded by Lemma (4.1). We bound  $\|\mathcal{T}^{-1}A_F\|_2$  by Lemma (??),

$$\|\mathcal{T}^{-1}A_F\|_2 \leq \|\mathcal{T}^{-1}\|_2 \|A_F\|_2 = \sigma_{\max}(\mathcal{T}^{-1}) \sigma_{\max}(A_F) = \frac{\sigma_{\max}(A_F)}{\lambda_{\min}(\mathcal{T})}.$$

Since  $A_F$  are small-norm this value is small. Using Lemma (4) we bound  $\|\mathcal{T}^{-1}\tilde{A}_E\|_2$ :

$$\lambda_k(\mathcal{T}^{-1}\tilde{A}_E) \leq \lambda_k(\tilde{A}_E) \frac{1}{\lambda_{\min}(\mathcal{T})} = \lambda_k(\mathcal{T}^{-1}\tilde{A}_E) \min_i \left( \tilde{\ell}_i(z_{\min}) \right) \leq \lambda_k(\mathcal{T}^{-1}\tilde{A}_E) \frac{1}{\varepsilon}. \quad (9)$$

To finish the proof, we utilize the fact that the matrices are SPD and relate the  $k$ -th eigenvalue of the preconditioned matrices to their 2-norm,

$$\begin{aligned} \|\mathcal{T}^{-1}A_{SN}\|_2 &\leq \|\mathcal{T}^{-1}H_{SN}\|_2 + \|\mathcal{T}^{-1}\tilde{A}_E\|_2 + \|\mathcal{T}^{-1}A_F\|_2 \\ &= \lambda_{\max}(\mathcal{T}^{-1}H_{SN}) + \lambda_{\max}(\mathcal{T}^{-1}\tilde{A}_E) + \lambda_{\max}(\mathcal{T}^{-1}A_F) \\ &\leq \frac{1}{\lambda_{\min}(\mathcal{T})} \left( \lambda_{\max}(H_{SN}) + \lambda_{\max}(\tilde{A}_E) + \lambda_{\max}(A_F) \right). \end{aligned}$$

The strength of this bound depends on  $\lambda_{\min}(\mathcal{T})$ , but can be written independently of  $\mathcal{T}$  in terms of the generating functions as in Equation (9). This dependence is the same for the single block case, and appears in similar proofs for discrete-trigonometric preconditioners (see for example [8]).

In practice the block size,  $m_i$ , is determined by the AFEM algorithm, which in turn dictates the actual size  $\varepsilon$  and hence the strength of clustering. So bigger Toeplitz blocks in  $A$  result in tighter eigenvalue clustering.

### 4.3 preconditioner LU

Using Schur complement's, we can obtain a block-LU decomposition for our permuted stiffness matrix. It will be notationally convenient to define the Schur complement matrix  $B := A_{22} - A_{21}A_{11}^{-1}A_{12}$ .

$$A = LU = \begin{pmatrix} I & 0 \\ A_{21}A_{11}^{-1} & I \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ 0 & B \end{pmatrix} \quad (10)$$

It is straightforward to verify the inverses of the two factor matrices are,

$$\begin{aligned} L^{-1} &= \begin{pmatrix} I & 0 \\ -A_{21}A_{11}^{-1} & I \end{pmatrix} \\ U^{-1} &= \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}B^{-1} \\ 0 & B^{-1} \end{pmatrix} \end{aligned}$$

Recalling that the size of  $A_{11}$  is the number of  $1 \times 1$  blocks, which is smaller and slower-growing than the dimension  $n$ , we can compute  $A_{11}^{-1}$  directly, without adding to our overall cost. We can invert  $B$  using the Sherman-Woodbury-Morris formula,  $B^{-1} = A_{22}^{-1} + A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}A_{12}A_{22}^{-1}$ . Notice, however, it requires the prohibitively expensive inversion of  $A_{22}$ . We know  $\mathcal{T}^{-1}$  is not only fast to compute, but also a good spectral approximation to  $A_{22}^{-1}$ . Let  $\hat{B}$  denote  $\mathcal{T} - A_{21}A_{11}^{-1}A_{12}$ . Now we can define

$$\hat{U} := \begin{pmatrix} A_{11} & A_{12} \\ 0 & \hat{B} \end{pmatrix}, \quad \hat{U}^{-1} := \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}\hat{B}^{-1} \\ 0 & \hat{B}^{-1} \end{pmatrix}.$$

Finally, our preconditioner is given by

$$\begin{aligned} M^{-1} &:= \hat{U}^{-1}L^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}\hat{B}^{-1} \\ 0 & \hat{B}^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -A_{21}A_{11}^{-1} & I \end{pmatrix} \\ &= \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}\hat{B}^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}\hat{B}^{-1} \\ -\hat{B}^{-1}A_{21}A_{11}^{-1} & \hat{B}^{-1} \end{pmatrix}. \end{aligned}$$

#### 4.4 final clustering

Theorem about clustering

Assumption  $\mathcal{T}^{-1}A_{22} = I + \varepsilon + N$ ,  $N$  is low-rank IE outliers

implies  $\mathcal{T} - A_{22} = \mathcal{T}\mathcal{T}^{-1}(\mathcal{T} - A_{22}) = \mathcal{T}(I - I - \varepsilon - N) = \mathcal{T}(-\varepsilon - N)$ .

$$\begin{aligned} M^{-1}A &= \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}\hat{B}^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}\hat{B}^{-1} \\ -\hat{B}^{-1}A_{21}A_{11}^{-1} & \hat{B}^{-1} \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \\ &= \begin{pmatrix} I + A_{11}^{-1}A_{12}\hat{B}^{-1}A_{21} - A_{11}^{-1}A_{12}\hat{B}^{-1}A_{21} & A_{11}^{-1}A_{12} + A_{11}^{-1}A_{12}\hat{B}^{-1}A_{21}A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12}\hat{B}^{-1}A_{22} \\ -\hat{B}^{-1}A_{21} + \hat{B}^{-1}A_{21} & -\hat{B}^{-1}A_{21}A_{11}^{-1}A_{12} + \hat{B}^{-1}A_{22} \end{pmatrix} \\ &= \begin{pmatrix} I & A_{11}^{-1}A_{12}\hat{B}^{-1}(\mathcal{T} - A_{22}) \\ 0 & \hat{B}^{-1}(A_{22} - A_{21}A_{11}^{-1}A_{12}) \end{pmatrix} \end{aligned}$$

Examining term in 12 spot:

$$\begin{aligned} A_{11}^{-1}A_{12}\hat{B}^{-1}(\mathcal{T} - A_{22}) &= A_{11}^{-1}A_{12}(\mathcal{T}^{-1} + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1})(\mathcal{T} - A_{22}) \\ &= A_{11}^{-1}A_{12}(I + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12})\mathcal{T}^{-1}(\mathcal{T} - A_{22}) \\ &= A_{11}^{-1}A_{12}(I + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12})(-\varepsilon - N) \end{aligned}$$

Examining term in 22 spot (basically following proof steps of SMW):

$$\begin{aligned}
\hat{B}^{-1}(A_{22} - A_{21}A_{11}^{-1}A_{12}) &= (\mathcal{T}^{-1} + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1})(A_{22} - A_{21}A_{11}^{-1}A_{12}) \\
&= \mathcal{T}^{-1}A_{22} + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1}A_{22} - \mathcal{T}^{-1}A_{21}A_{11}^{-1}A_{12} - \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1}A_{21}A_{11}^{-1}A_{12} \\
&= I + \varepsilon + N [\mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}(I + \varepsilon + N) - \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1}A_{21}A_{11}^{-1}A_{12}] \\
&= I + \varepsilon + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}(\varepsilon + N) + [\mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1}A_{21}A_{11}^{-1}A_{12}] \\
&= I + \varepsilon + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}(\varepsilon + N) + [-\mathcal{T}^{-1}A_{21}A_{11}^{-1} + (\mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1}A_{21}A_{11}^{-1})]A_{12} \\
&= I + \varepsilon + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}(\varepsilon + N) + [-\mathcal{T}^{-1}A_{21}A_{11}^{-1} + (\mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}\mathcal{T}^{-1}A_{21}A_{11}^{-1})]A_{12} \\
&= I + \varepsilon + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}(\varepsilon + N) + [-\mathcal{T}^{-1}A_{21}A_{11}^{-1} + \mathcal{T}^{-1}A_{21}A_{11}^{-1}]A_{12} \\
&= I + \varepsilon + \mathcal{T}^{-1}A_{21}(A_{11} - A_{12}\mathcal{T}^{-1}A_{21})^{-1}A_{12}(\varepsilon + N).
\end{aligned}$$

## 5 Numerical Results

### 5.1 Validating Assumptions

In order to prove the spectral clustering, we made two assumptions about the structure of  $A$ : the blocks adjacent to the diagonal are well-approximated by low-rank matrices and the other off-diagonal blocks are small-norm. Table (1) displays the maximum and average (median) rank needed to achieve an approximation under  $\varepsilon$  on each block. Table (2) shows the norm and relative norm of  $A_F$ . Both tables include results for  $\alpha = 1.3, 1.5, 1.9$ , after 7 levels of refinement, but similar trends were seen across all fractional orders and refinement levels. In the case of  $1 \times 1$  Toeplitz blocks we found that including the entry to the right and underneath in  $A_E$  was not sufficient to capture all the low-rank information. Instead, the corresponding block in  $A_E$  was extended to dimensions  $5 \times 5$  such that the original entry is the center of that block. In terms of spectral clustering, this could increase the number of outliers, but better characterizes the size of the cluster.

| $\varepsilon \backslash \alpha$ | 1.3  | 1.5  | 1.9  |
|---------------------------------|------|------|------|
| $10^{-3}$                       | 4(5) | 4(6) | 5(6) |
| $10^{-4}$                       | 5(7) | 5(7) | 6(8) |
| $10^{-5}$                       | 6(8) | 6(9) | 7(9) |

Table 1: The approximation rank needed for the error to be less than  $\varepsilon$  for varying fractional indices  $\alpha$ . Table entries are displayed as **median (max)**.

| $\alpha$                    | 1.3                    | 1.5                    | 1.9                    |
|-----------------------------|------------------------|------------------------|------------------------|
| $\ A_F\ _2$                 | 0.6138                 | 1.528                  | 4.479                  |
| $\frac{\ A_F\ _2}{\ A\ _2}$ | $4.893 \times 10^{-2}$ | $2.368 \times 10^{-2}$ | $2.533 \times 10^{-3}$ |

Table 2: The norm of  $\|A_F\|_2$  and  $\|A_F\|_2/\|A\|_2$  across values of  $\alpha$ .

## 5.2 Improved Convergence

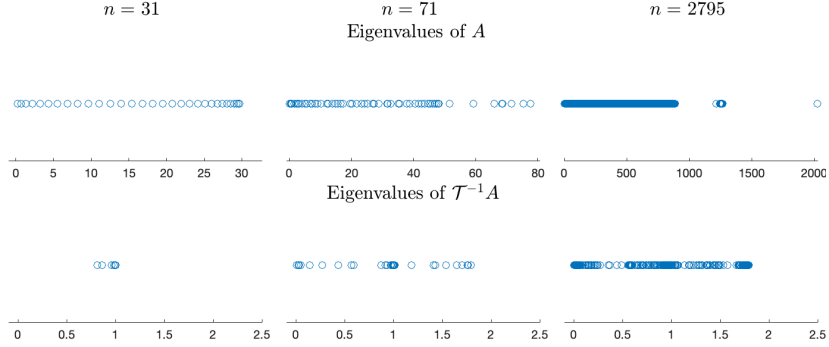


Figure 2: Eigenvalue clustering before and after preconditioning. Each column represent a different matrix size. The first row shows the spectrum before preconditioning, the second shows the spectrum, on a smaller interval, after preconditioning.

Example results of the spectrum before and after preconditioning are in Figure (2). These tests are for the problem (1) with  $\alpha = 1.7$  and exact solution  $u(x) = 10x^2(1 - x^2)$  on  $[0, 1]$  (see [13] example 6.1 for more details). The three matrix sizes are  $n = 31, 71, 2795$  and reflect 0, 3, and 11 levels of refinement respectively. Note that zero levels of refinement is a uniform mesh.

The left-most column represents a matrix that consists of a single Toeplitz block, so the preconditioner reduces to the case of  $\tau$ . In this case the clustering of the spectrum around 1 is very strong. In the  $n = 71$  case, the biggest Toeplitz blocks are size 11, and there are two such blocks. The eigenvalue clustering around 1 is still observable, but the size of the cluster has grown. Finally for  $n = 2795$ , the preconditioner reduces the spectral radius by several orders of magnitude. The spectrum is concentrated around 1, but the distribution of eigenvalues between 0 and 2 is more uniform. For this matrix, the biggest Toeplitz block is size 939 and there are two other blocks just bigger than 200. On the other hand, there are 245  $1 \times 1$  blocks. For all cases, the strength of the clustering is explained by the amount of smaller Toeplitz blocks in  $A$ . Notice that although the proof allows for eigenvalue outliers beyond the cluster, in practice everything remains within the interval  $(0, 2)$ .

Next, we see how the clustered spectrum translates to accelerated convergence. In Figure (3) the same matrices are used for preconditioned conjugate gradient (PCG), comparing the adaptive  $\tau$  preconditioner and a Jacobi preconditioner with a tolerance of  $10^{-8}$ . For the two smaller matrices, the preconditioner clearly accelerates the point of superlinear convergence [11]. For the final matrix, the Jacobi-preconditioned system fails to converge within the maximum of 600 iterations, but with the  $\mathcal{T}$  preconditioner converges in about 300 iterations.

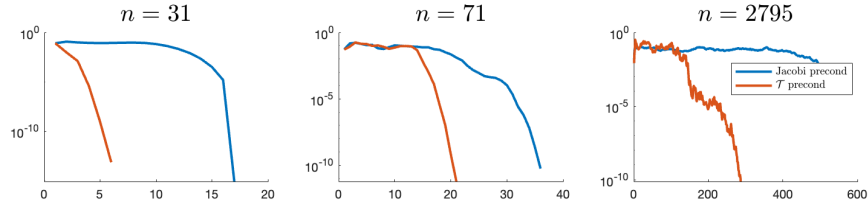


Figure 3: Comparing the residual vector norm across PCG iterations. The blue line uses a Jacobi preconditioner and the orange line uses the  $\mathcal{T}$  preconditioner.

## 6 Conclusion

We have shown how to build a preconditioner for FDEs discretized by AFEM. The spectrum of the preconditioned system is clustered around 1 and accelerates convergence of iterative methods. In the future, we will investigate how to customize the AFEM MARK module to build bigger Toeplitz blocks and thus get tighter clusters. Finally, we will investigate how to use these techniques to build preconditioners for FDEs discretized with AFEM on higher dimensional meshes.

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